

Lecture 16:

Smooth Fictitious Play and the Optimization Premium

1. Smooth fictitious play combines probabilistic choice and assessments based on the historical frequency of play.
 - a. Specifically, smooth fictitious play assigns to each player i a smooth response function, $\bar{BR}^i$, like the logit response function,

$$\bar{BR}_i(\sigma_{-i})(s_i) = \frac{e^{\lambda u_i(s_i, \sigma_{-i})}}{\sum_{r_i \in S_i} e^{\lambda u_i(r_i, \sigma_{-i})}}$$

- b. and an assessment rule, γ_t^i , where the assessment rule is the same used in fictitious play:
 - i. Player i has an exogenous initial weight function $\kappa_0^i: S^{-i} \rightarrow \mathbb{R}_+$
 - ii. The weights are updated by adding 1 to the weight of each opponent strategy each time it is played

$$\kappa_t^i(s^{-i}) = \kappa_{t-1}^i(s^{-i}) + \begin{cases} 1 & \text{if } s_{t-1}^{-i} = s^{-i}, \\ 0 & \text{if } s_{t-1}^{-i} \neq s^{-i} \end{cases}$$

- iii. The probability that player i assigns to player $-i$ playing s^{-i} at date t is given by

$$\gamma_t^i(s^{-i}) = \frac{\kappa_t^i(s^{-i})}{\sum_{\tilde{s}^{-i} \in S^{-i}} \kappa_t^i(\tilde{s}^{-i})}$$

2. Fudenberg and Kreps (1993) prove that the logit equilibrium corresponding to a mixed strategy equilibrium is globally stable in 2×2 games with a unique mixed equilibrium when λ is sufficiently small,
 - a. So smooth fictitious play can provide an explanation of how learning can lead agents' play in each period to correspond to a logit equilibrium.
 - b. Although, notice that the restriction on the size of λ may result in the logit equilibrium being far from the mixed Nash equilibrium it converges to when λ goes to infinity.
3. Benaim and Hirsch (1996) prove that
 - a. smooth fictitious play converges to the logit equilibrium in 2×2 games with a unique strict equilibrium
 - b. And that the logit equilibrium corresponding to the mixed equilibrium in 2×2 games with two strict and one mixed equilibrium is unstable.
 - c. They give a three player example of a game with a unique equilibrium in which smooth fictitious play can get trapped in cycles.
4. Chen, Friedman, and Thisse (1997) give sufficient conditions for convergence to the logit equilibrium for finite n -person games, which involve restrictions on the size of λ such that the logit response functions form a contraction mapping.

5. Fudenberg and Kreps prove convergence using stochastic approximation methods.
 - a. The long-run behavior of discrete-time time-averaging stochastic systems can be determined by analyzing a related deterministic system in continuous time.
 - b. Specifically, they show that asymptotically the dynamics resembles that of the continuous-time near-best-response dynamics:

$$\dot{\theta}^i = \bar{BR}^i(\theta) - \theta^i$$

where θ^i is the probability distribution over strategies by types assigned to player role i .

- c. If a point or cycle is an unstable orbit of the continuous-time dynamic, the noise in the system will knock it into a stable orbit.
 - d. So the stochastic system can only converge to stable orbits of the continuous-time system.

	X	Y
X	45,45	0,35
Y	35,0	40,40

$2R$ Game Form

	X	Y
X	45,45	0,40
Y	40,0	20,20

R Game Form

	X	Y
X	45,45	0,42
Y	42,0	12,12

$0.6R$ Game Form

6. Under the assumption that players maximize their earnings, each game has two pure-strategy equilibria:
 - a. the payoff-dominant equilibrium (X,X)
 - b. the risk-dominant equilibrium (Y,Y) .
 - c. Each game also has a mixed equilibrium in which X is played with probability k^* , where k^* equals 0.8.
7. The best response correspondence in games $2R$, R and $0.6R$ is completely determined by k^* .
 - a. Strategy X is a strict best response to any mixture that attaches a probability greater than k^* to X ,
 - b. while Y is a strict best response to any mixture attaching a lower probability to X .
8. To the extent possible, the games also involve payoffs of similar magnitudes. In particular, the expected payoff from the mixed equilibrium is 36 for all three games.

OPTIMIZATION PREMIUM

1. Let p denote the probability a player chooses X and q denote the probability his opponent chooses X .
2. Then the expected payoff for the player is

$$\pi_M(p, q) = \{p, 1-p\} \cdot M \cdot \{q, 1-q\},$$

where M denotes the relevant earnings matrix.

3. The earnings difference, given mixture k , is given by

$$\begin{aligned} r_{2R}(k) &= \pi_{2R}(X, k) - \pi_{2R}(Y, k) = 50(k - k^*) \\ r_R(k) &= \pi_R(X, k) - \pi_R(Y, k) = 25(k - k^*) \\ r_{0.6R}(k) &= \pi_{0.6R}(X, k) - \pi_{0.6R}(Y, k) = 15(k - k^*). \end{aligned}$$

4. Hence, for any mixture k , the earnings difference between the two actions is twice as large in game $2R$ as it is in game R and six tenths as large in game $0.6R$ as it is in game R .

SELECTION THEORIES

1. Classical selection theories treat games $2R$, R and $0.6R$ identically.
 - a. Carlsson and van Damme (1993) and Harsanyi (1995) choose the risk-dominant equilibrium.
 - b. Anderlini (1990) and Harsanyi and Selten (1988) choose the payoff-dominant equilibrium.
2. Deterministic selection dynamics predict history dependent equilibrium selection.
 - a. The replicator dynamic or myopic best response dynamic, choose either the payoff-dominant or risk-dominant equilibrium, depending upon which basin of attraction contains the initial condition.
 - b. The three games have the same separatrix, k^* , under either the replicator or myopic best response dynamic.
 - c. If the initial state is on the same side of k^* , the dynamic picks the same equilibrium in all three games.

3. Stochastic selection dynamics
 - a. Kandori, Mailath and Rob (1993) and Young (1993) select the risk-dominant equilibrium.
 - b. Robson and Vega-Redondo (1996) examine a similar model in which the payoff-dominant equilibrium is selected.
 - c. Despite their different outcomes, each of these theories selects the same equilibrium for all three games in the ultra-long run.
4. The limiting logit equilibrium selects the risk-dominant equilibrium for all three games.

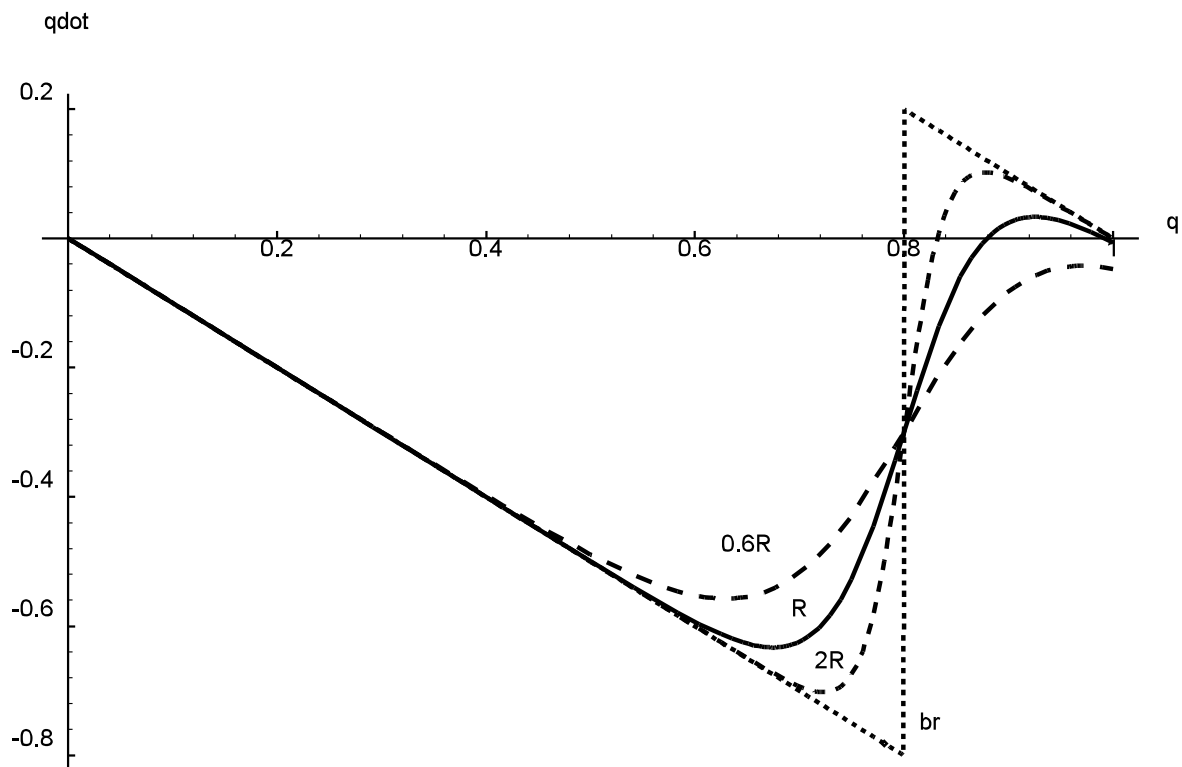


Figure 4: One population continuous time best response and logit response dynamics ($\lambda=1$).

$$r(q;M) \equiv u(X,q;M) - u(Y,q;M) \equiv \{1,-1\} \cdot M \cdot \{q,1-q\}.$$

$$p(q,\lambda;M) = \frac{\exp(\lambda r(q;M))}{1 + \exp(\lambda r(q;M))}.$$

$$\dot{q} = p(q,\lambda;M) - q,$$

REPLICATOR DYNAMIC

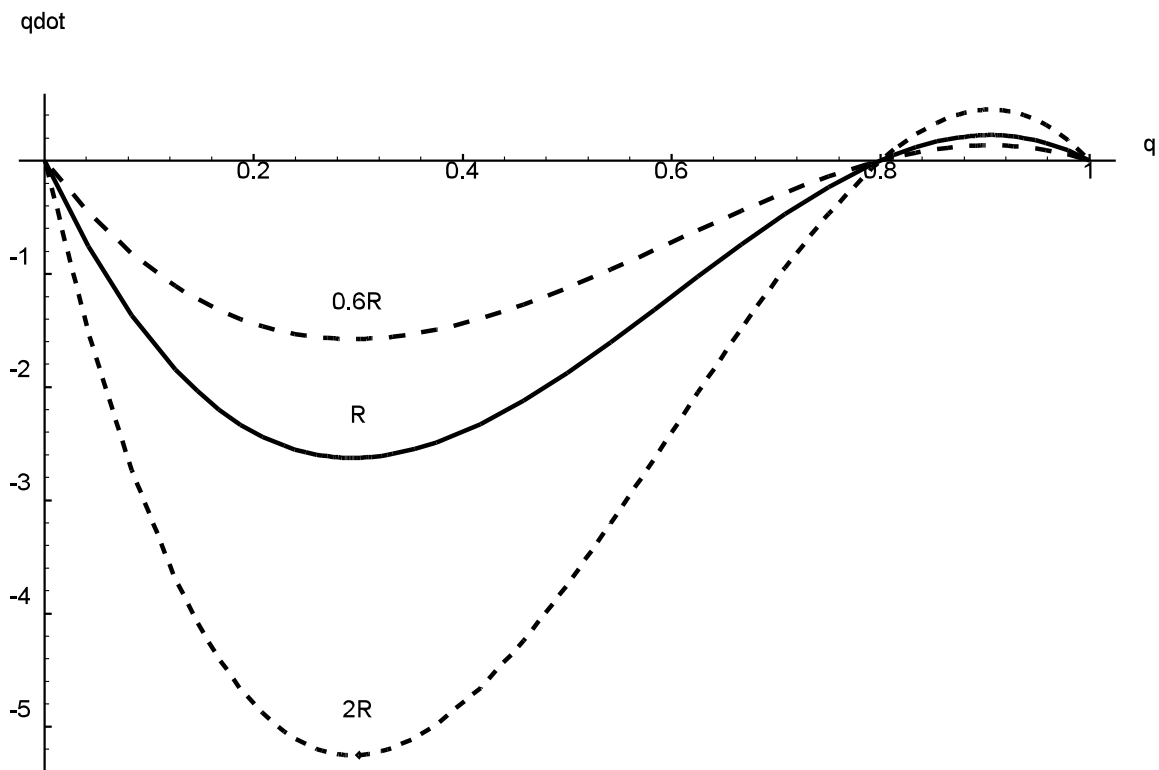


Figure 5: Continuous Time Replicator Dynamic for game *RD* (dashed line) and *PD* (solid line).

$$\dot{q} = q(\{1,0\}.M.\{q,1-q\} - \{q,1-q\}.M.\{q,1-q\}),$$

$$\begin{aligned} \frac{\dot{q}}{q}|_{2R} &= 2 \frac{\dot{q}}{q}|_R \\ \frac{\dot{q}}{q}|_{0.6R} &= 0.6 \frac{\dot{q}}{q}|_R. \end{aligned}$$

Hypothesis 1: Behavior will converge to a mutual best response outcome more quickly the larger is the optimization premium.

Hypothesis 2: Behavior is more likely to cross the separatrix and converge to the payoff-dominant equilibrium the smaller is the optimization premium.

Hypothesis 3: Subjects' behavior will be closer to best-response behavior the larger is the optimization premium.

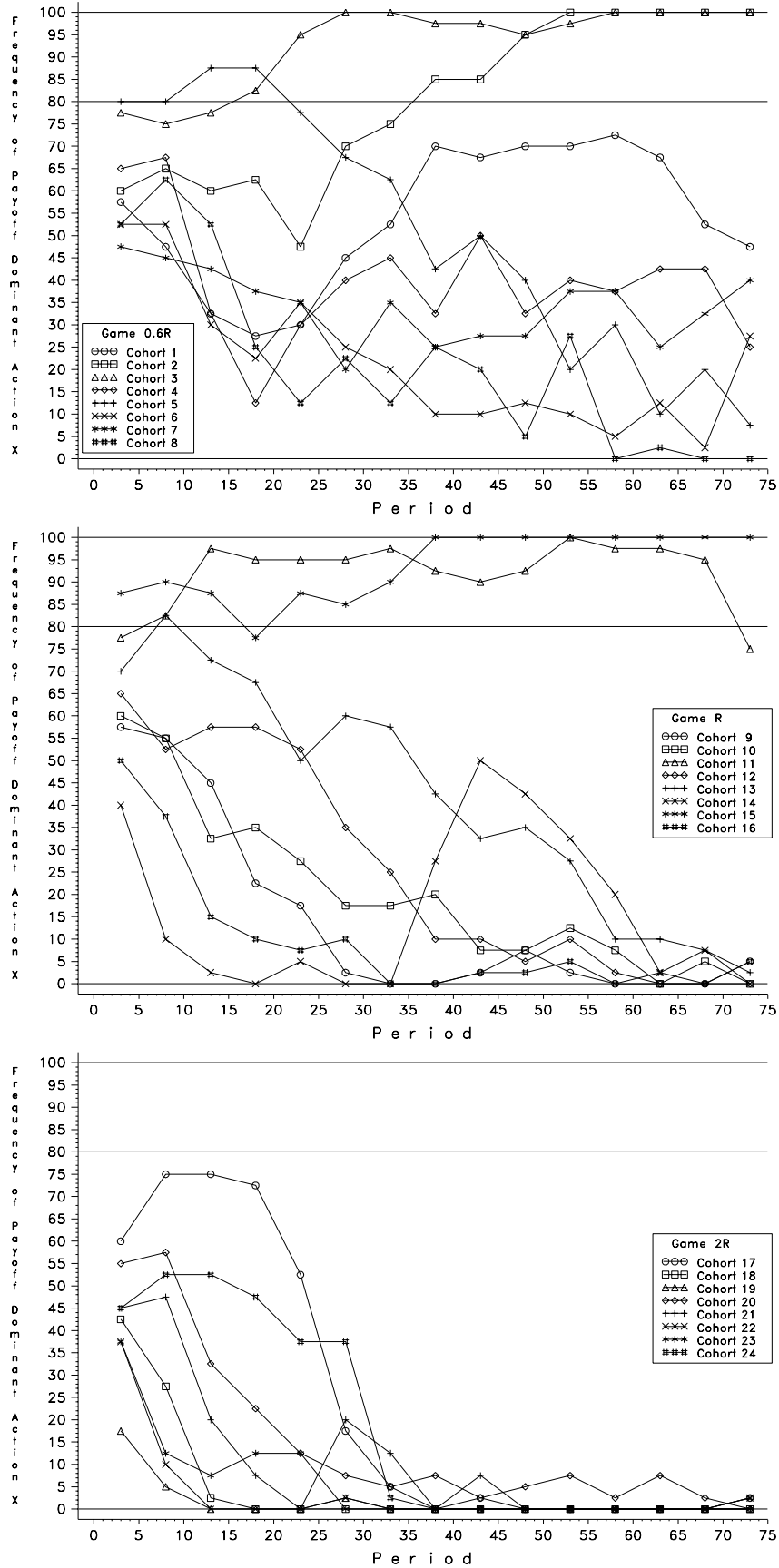


Figure 6: Five period mean frequency of X by cohort.

EXPERIMENTAL RESULTS

1. In period 1, 62.5 percent of the subjects play action X , the payoff-dominant action.
 - a. Risk-dominance is not a salient deductive selection principle.
 - b. The modal subject thus plays the payoff-dominant action in the first period,
 - c. but not enough subjects focus on payoff dominance to make playing the payoff-dominant action a best response, since 0.625, or 5 of 8, is less than k^* .

Table 4: Ranking Overall frequency of X by Cohort

Game	Cohort	Initial Outcome	Terminal Outcome	Overall Frequency of X	Rank	Average Per Capita Earnings
$0.6R$	1	0.500	0.375	0.540	20	\$19.67
$0.6R$	2	0.750	1.000	0.803	21	\$27.35
$0.6R$	3	0.750	1.000	0.930	23	\$31.31
$0.6R$	4	0.750	0.250	0.397	17	\$16.31
$0.6R$	5	0.750	0.125	0.508	19	\$19.39
$0.6R$	6	0.500	0.375	0.218	13	\$12.59
$0.6R$	7	0.625	0.375	0.343	16	\$14.76
$0.6R$	8	0.500	0.000	0.213	12	\$12.86
R	9	0.750	0.125	0.145	7	\$15.88
R	10	0.625	0.000	0.203	11	\$16.12
R	11	0.750	0.625	0.920	22	\$31.00
R	12	0.750	0.000	0.255	15	\$17.06
R	13	0.750	0.125	0.418	18	\$19.44
R	14	0.500	0.000	0.160	9	\$15.63
R	15	0.750	1.000	0.937	24	\$31.44
R	16	0.750	0.125	0.098	5	\$15.38
$2R$	17	0.625	0.125	0.242	14	\$27.22
$2R$	18	0.500	0.000	0.048	3	\$28.99
$2R$	19	0.375	0.000	0.017	1	\$29.44
$2R$	20	0.625	0.000	0.152	8	\$27.01
$2R$	21	0.375	0.000	0.107	6	\$28.03
$2R$	22	0.500	0.125	0.035	2	\$29.19
$2R$	23	0.750	0.000	0.055	4	\$28.51
$2R$	24	0.500	0.125	0.185	10	\$26.63

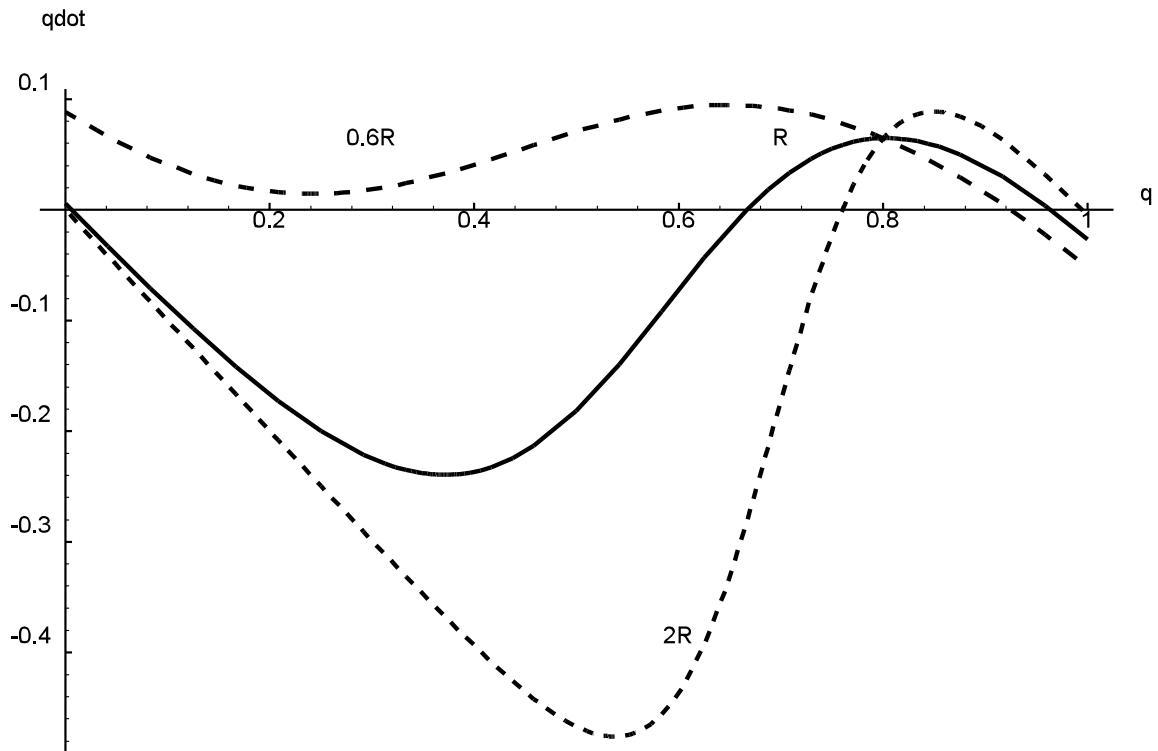


Figure 7: Logit response dynamic based on estimated EFP model.

CONVERGENCE

1. If we examine the first (five period) state in which every subject in a cohort plays the risk-dominant action (excluding the R cohorts that never converge to the risk-dominant equilibrium),
 - a. we find that the remaining six R cohorts take an average of 50 periods for all subjects to reach the risk-dominant equilibrium,
 - b. while the eight $2R$ cohorts take an average of 26 periods.
 - c. Only one $0.6R$ cohort converges to the risk-dominant equilibrium state. It took 58 periods.
2. The evidence is thus consistent with the prediction that reducing the optimization premium reduces the speed of convergence to the inefficient risk dominant equilibrium.

SUMMARY

1. Subjects don't bring risk dominance into the laboratory, but risk dominance makes a more accurate *a priori* prediction of the terminal outcome than payoff dominance or security.
2. The optimization premium has a statistically and economically significant effect on the observed frequency of the payoff dominant action.

Table 5: Recent Evidence on Human Behavior in Evolutionary Stag Hunt Games.

N	Game = {A, C} {B, D}	k^*	$R(k)/(k-k^*)$	Initial Outcome	Terminal Outcome	Periods	Source
1	{100,0},{90,70} ^T	0.87	4/5	13/20	2/20	10	Clark, <i>et al.</i>
2	{100,0},{90,70} ^T	0.87	4/5	8/20	2/20	10	Clark, <i>et al.</i>
3	{100,0},{80,80} ^T	0.80	1	6/10	0/10	22	Cooper, <i>et al.</i> (1992)
4	{100,0},{80,80} ^T	0.80	1	6/10	0/10	22	Cooper, <i>et al.</i> (1992)
5	{100,0},{80,80} ^T	0.80	1	5/10	1/10	22	Cooper, <i>et al.</i> (1992)
6	{100,0},{80,80} ^T	0.80	1	7/20	4/20	10	Clark, <i>et al.</i> (1996)
7	{100,0},{80,80} ^T	0.80	1	5/20	2/20	10	Clark, <i>et al.</i> (1996)
8	{100,0},{80,80} ^T	0.80	1	4/10	0/10	9	Straub (1995)
9	{100,0},{70,90} ^T	0.75	6/5	10/20	0/20	10	Clark, <i>et al.</i>
10	{100,0},{80,60} ^T	0.75	4/5	4/10	0/10	8	Schmidt, <i>et al.</i>
11	{100,0},{80,60} ^T	0.75	4/5	5/10	0/10	8	Schmidt, <i>et al.</i>
12	{100,0},{80,60} ^T	0.75	4/5	5/10	4/10	8	Schmidt, <i>et al.</i>
13	{100,0},{80,60} ^T	0.75	4/5	5/10	5/10	8	Schmidt, <i>et al.</i>
14	{100,20},{80,80} ^T	0.75	4/5	2/10	0/10	9	Straub (1995)
15	{100,20},{80,80} ^T	0.75	4/5	4/10	1/10	8	Schmidt, <i>et al.</i>
16	{100,20},{80,80} ^T	0.75	4/5	5/10	7/10	8	Schmidt, <i>et al.</i>
17	{100,20},{80,80} ^T	0.75	4/5	7/10	8/10*	8	Schmidt, <i>et al.</i>
18	{100,20},{80,80} ^T	0.75	4/5	7/10	10/10*	8	Schmidt, <i>et al.</i>
19	{5,-1},{3,3}	0.67	6/5	5/12	3/12	10	Friedman (1996)
20	{5,-1},{4,1}	0.67	3/5	7/12	3/12	10	Friedman (1996)
21	{5,-1},{4,1}	0.67	3/5	7/12	3/12	16	Friedman (1996)
22	{5,-1},{4,1}	0.67	3/5	6/10	3/10	16	Friedman (1996)
23	{5,-1},{4,1}	0.67	3/5	2/12	3/12	10	Friedman (1996)
24	{5,-1},{4,1}	0.67	3/5	3/12	2/12	10	Friedman (1996)
25	{80,10},{70,30} ^T	0.67	3/8	9/10	9/10	9	Straub (1995)
26	{100,20},{60,60} ^T	0.50	4/5	7/10	10/10	9	Straub (1995)
27	{100,20},{60,60} ^T	0.50	4/5	9/10	9/10	8	Schmidt, <i>et al.</i>
28	{100,20},{60,60} ^T	0.50	4/5	9/10	9/10	8	Schmidt, <i>et al.</i>
29	{100,20},{60,60} ^T	0.50	4/5	3/10	7/10*	8	Schmidt, <i>et al.</i>
30	{100,20},{60,60} ^T	0.50	4/5	5/10	9/10	8	Schmidt, <i>et al.</i>
31	{5,0},{4,1}	0.50	2/5	6/12	9/12	10	Friedman (1996)
32	{100,20},{80,80} ^T	0.50	2/5	9/10	10/10	8	Schmidt, <i>et al.</i>
33	{100,20},{80,80} ^T	0.50	2/5	8/10	10/10	8	Schmidt, <i>et al.</i>
34	{100,20},{80,80} ^T	0.50	2/5	7/10	9/10	8	Schmidt, <i>et al.</i>
35	{100,20},{80,80} ^T	0.50	2/5	6/10	8/10	8	Schmidt, <i>et al.</i>
36	{55,25},{35,35} ^T	0.33	6/11	9/10	10/10	9	Staub (1995)

$R(k)$ - scaled earnings difference given k , the probability of X : $R(k) = (k A + (1-k) C) - (k B + (1-k) D) / A$

k^* - separatrix, zero earnings difference, mixed strategy equilibrium.

T - payoff dominant equilibrium in the lower right cell of subjects earnings table in original experiment.

* - Separatrix crossings between initial and terminal outcome.

3. A convention emerges.
 - a. The inefficient equilibrium emerges when $k^* > 0.75$.
 - b. The efficient equilibrium emerges when $k^* < 0.5$.
 - c. For $0.5 < k^* < 0.75$ results are mixed.
4. Violations of the separatrix are rare and depend on the scaled earnings difference.
 - a. The terminal outcome is accurately predicted by the location of the initial outcome in the respective equilibrium's basin of attraction.
 - b. The scaled payoff difference between the two actions influences both the initial outcome and the frequency of observed violations of the separatrix.
5. Our estimated basins of attraction are different from the predictions of the best-response or replicator dynamic, but not in the way predicted by the analysis of logit equilibria.